

Scientific Advisory Committee on the Medical Implications of Less-Lethal Weapons (SACMILL)

Statement on the Medical Implications of the TASER 10™ Conducted Energy Device System

Key points:

- The TASER 10™ is a new type of Conducted Energy Device (CED) the concept of use of which marks a major departure from other CED types that have been authorised for use in the UK (TASER M26™, TASER X26™, TASER X2™ and TASER 7™). Previous CEDs fired up to two pairs of probes of opposite polarity, with each pair being propelled on divergent trajectories from the device following a single trigger activation. The divergence is needed to create the probe spread required for the electrical discharge to induce neuromuscular incapacitation.
- The TASER 10™, in contrast, fires up to ten individually targeted probes, requiring up to ten trigger activations. A minimum of two probes must be in contact with an individual for incapacitation to be induced. Unlike earlier models, however, the darts of the TASER 10™ probes must embed in the skin for incapacitation to occur. Moreover, because the probes are fired individually, it is for the user to create the probe spread necessary for incapacitation.
- Whereas the polarities of the probe pairs fired from earlier devices were fixed, each probe of the TASER 10™ can assume either a positive or negative polarity. Therefore, when discharging probes at an individual this means that, starting with a full magazine, there are forty-five prospective probe pairings available with which to obtain two skin-embedded darts. At the other end of the scale, should the first eight probes fail to embed in the skin, then only a single opportunity remains to make an electrical circuit with the final two probes.
- The range of the TASER 10™ probes exceeds that of probes fired from previous devices, with the capacity to reach up to 13.7 m (45 ft) and beyond. The enhanced range may give officers more time and distance in which to use de-escalation tactics before resorting to firing probes.
- When fired, the kinetic energy (KE) of TASER 10™ probes exceeds that of probes fired from the earlier devices: when measured close to their point of exit from the device, the KE of TASER 10™ probes is some 18% higher than TASER 7™ probes and 118% higher than TASER X2™ probes. This serves to increase the maximum engagement range of the TASER 10™ and is said to improve the likelihood of obtaining skin-embedded darts. On the other hand, the increased probe energy also means that the dispersal of kinetic energy on impact may be more extensive and dart penetration injuries may be more serious than those from earlier generations of probe.

- Unlike previous models, the TASER 10™ cannot generate a high voltage warning arc display due to the absence of a brief very high voltage component from the start of its output pulse. For the same reason, the TASER 10™ cannot arc across a small air gap to traverse clothing nor can it be used to directly apply a painful shock in contact mode (commonly called ‘drive-stun’ and ‘angled drive-stun’). The TASER 10™ replaces the arc display with an audio-visual warning intended to aid de-escalation. The efficacy of this audio-visual warning is unknown and, therefore, could be a potential weakness in the new device’s deterrent capability. The absence of ‘drive-stun’ on the TASER 10™ removes one use of force option that may have held value for users of earlier devices.
- Barring the absence of a brief very high voltage component, the electrical output of the TASER 10™ is similar to that of earlier CEDs, with neuromuscular incapacitation being achieved through the administration of repetitive pulses, with each pulse resulting from current flow between a single pair of probes. This remains the case even though the subject may have three or more TASER 10™ probes embedded in their skin, although the identity and polarity of the probe pairs contributing to a pulse may change during the course of the discharge cycle.
- The medical implications of the foregoing characteristics of the TASER 10™ are among those considered by SACMILL in this statement, set in the wider context of the TASER 10™ system as a whole. Also considered are the findings from technical testing, the performance of the device in user handling trials, human factors elements, the proposed national training curriculum and associated qualification testing, how the roll-out of the CED would be managed and how the operational performance of the TASER 10™ system would be monitored should it be authorised by the Home Office.
- Should the system be authorised, it is vitally important that any significant deviations from the medical implications articulated in this statement are reflected in a revised statement. Furthermore, should any substantive element of the system be changed, SACMILL must be informed and any consequential medical implications considered.
- Should the TASER 10™ system be authorised by the Home Office, it should be mandatory that the operational performance in the hands of the end-user is monitored to obtain reassurance that it performs in the manner anticipated. This monitoring should continue for a minimum of twelve months and SACMILL must be informed immediately of any adverse, or potentially adverse, medical outcomes that may have a bearing on the advice expressed in this medical statement. SACMILL anticipates formal reporting on the operational performance of the TASER 10™ system within 18 months of its introduction into policing.

Preamble

1. SACMILL is a non-departmental public body that provides independent advice to Ministers of His Majesty's Government. This advice concerns the medical aspects surrounding the use by the police and other authorised bodies of less-lethal weapons (LLWs)¹ on members of the public.² SACMILL is sponsored by the Surgeon General in the Ministry of Defence (MoD).
2. SACMILL acts only in an advisory capacity and neither endorses nor approves the LLW systems under review.
3. In addressing its remit, SACMILL considers all aspects of a LLW system that may have a bearing on the equipment's operational use. These include: developing an understanding of the effects of the weapon's output on the human body; the content and quality of the user guidance and training; how the equipment will be stored and maintained; the manner in which the system will be deployed and used; monitoring and learning from adverse outcomes arising in operational use of LLWs in the UK and elsewhere; assessing the implications of basic research into the medical effects of LLWs; and recommending avenues for further research. SACMILL also considers what information should be made available to personnel involved in the medical management of people subjected to a given LLW system.
4. In recognising that the use of LLWs is not free of medical risk, SACMILL will seek to understand and articulate the risk by systematically evaluating all elements of a less-lethal system and advising Ministers and other stakeholders accordingly.
5. This medical statement, which has been prepared at the request of the Home Office, provides SACMILL's opinion on the medical implications of the TASER 10™ conducted energy device (CED) system.

Evidence reviewed by SACMILL

6. In forming a view on the medical implications of the TASER 10™ system, SACMILL has partly relied on the following evidence:
 - (a) A report by the Defence Science and Technology Laboratory (Dstl) entitled *Dstl opinion on the medical implications of the TASER 10™ system*. (Dstl/CR164019 v1.0, 21/02/2025)
 - (b) Slide presentations by Dstl at SACMILL meetings on 22/11/2024 and 16/12/2024.
 - (c) A report on the outcome of independent testing entitled: *Taser 10 Technical Testing Results* (16/01/2025; Ref: MIQ-24-0014-D version K; PA Consulting Services Ltd).
 - (d) *Assessment of TASER 10™: User handling trial*. Version 1.0 (October 2024), College of Policing.

¹ LLWs are sometimes referred to as non-lethal weapons or less-than-lethal weapons.

² The health and safety implications for users of less-lethal systems will be routinely assessed by law enforcement agencies. SACMILL will take these user-related aspects into account when forming a view on medical implications for the public.

- (e) TASER 10™ user training and assessment documentation from the College of Policing (received 18 March 2025).
- (f) Anecdotal evidence from international law enforcement agencies and non-peer reviewed operational reports.
- (g) A document from the National Police Chiefs' Council and College of Policing entitled *Implementation programme for TASER® 10™ Conducted Energy Device (CED)* (circulated 09/06/2025)
- (h) A document from the National Police Chiefs' Council entitled *TASER 10 Fact Sheet* (V3.0 circulated 23/06/2025)
- (i) *Taser 10 Technical Testing Results – Additional Testing* (12/03/2025; Ref: MIQ-24-0014-D_L2; PA Consulting Services Ltd).
- (j) *Taser 10 Technical Testing Results – Further Probe Synchronisation Testing* (30/06/2025; Ref: MIQ-25-0023-D_E; PA Consulting Services Ltd).
- (k) A letter report by the Defence Science and Technology Laboratory (Dstl) entitled *Additional testing of the TASER 10™*. (Dstl/LR170350 v1.0, 02/07/2025)

In addition to the above, SACMILL received a TASER 10™ familiarisation session from the College of Policing after which the Committee undertook live firing of the device.

(Note that evidence item 6(c) covered the outcome of the first round of technical testing of the TASER 10™, while items 6(i) and 6(j) covered the additional testing requested by SACMILL following its initial review of the TASER 10™ system. SACMILL has been advised that these three reports will be consolidated into a single report covering the full range of technical testing undertaken.)

Overview of the TASER 10™

7. The TASER 10™ was launched by Axon Enterprise, Inc, in January 2023. Earlier CEDs produced by the company include the TASER M26™ (authorised by the Home Secretary in 2003), the TASER X26™ (authorised in 2005), the TASER X2™ (in 2017) and the TASER 7™ (in 2020). The TASER M26™ is no longer in service in the UK while the TASER X26™ has all but been phased out in favour of the more recent TASER X2™ and TASER 7™ devices. SACMILL considered the medical implications of the TASER X2™ system in 2016 and the TASER 7™ system in 2020.^{3,4}
8. The TASER X2™ and TASER 7™ are twin cartridge devices that fire a pair of probes of opposite electrical polarity from each cartridge when the trigger is activated. The upper (positive) probe is ejected from the cartridge in line with the 'muzzle' of the device while the lower (negative) probe diverges downward at an angle set by the cartridge design. For neuromuscular incapacitation (NMI) to be induced, at least two probes of opposite polarity must make electrical contact with the subject and the probe spread on the subject's body must be sufficiently wide (hence, the diverging trajectories of the probes as they leave the cartridge). A minimum probe spread of 23-30 cm

³ <https://www.gov.uk/government/publications/medical-implications-of-the-taser-x2>

⁴ <https://www.gov.uk/government/publications/medical-statement-and-technical-reports-for-the-taser-7>

(9-12 inches) is required to consistently induce NMI.⁵

9. In common with other Axon CEDs, the TASER 10™ achieves NMI through the application of short, repetitive electrical pulses which are administered via the probes. The resulting NMI is understood to result from the stimulation of sensorimotor nerves in the vicinity of the probes and involves multiple spinal reflexes.⁶
10. In contrast to the earlier Axon devices, the TASER 10™ can discharge up to ten individually fired floating polarity probes, meaning that each probe can assume either a positive or negative polarity. In principle, then, with a fully loaded TASER 10™ there are forty-five opportunities to create the probe pairing required to induce NMI. The number of probe pairing possibilities progressively decreases as more probes are discharged and, should none of the darts of the first eight probes embed in the skin, then only one opportunity remains with the final two probes to create an electrical circuit.
11. As the probes of the TASER 10™ are fired individually rather than in pairs, it is for the firer to create the required probe separation on the subject by appropriate targeting of the probes. For earlier devices, the probe spread was determined by a combination of the cartridge design and the distance of the subject from the CED officer.
12. Another significant departure of the TASER 10™ from earlier Axon devices is the absence of a brief, very high voltage component at the start of the output waveform. On previous models, this component enabled the waveform to arc across a small air gap, which could occur if the probes landed in clothing.⁷ The operational implications of this, based on officer-reported data on the ability of the electrical discharge to subdue subjects, have been reported.⁸
13. The loss of the very high voltage component of the TASER 10™ waveform has three important corollaries. Firstly, for the TASER 10™ discharge to induce NMI, the darts of at least two of the probes fired from the device must embed in the skin of the subject with sufficient separation.⁹ Secondly, the TASER 10™ can no longer produce an arc warning display at the front of the device. Instead, the arc warning display of the earlier types of CED is replaced by a warning alert consisting of a high brightness strobing flashlight in combination with an electronically synthesised sound. Thirdly, there is no longer a 'contact mode' (formerly known as 'drive stun'), in which the front end of the device is brought into direct contact with the subject to impart a painful shock. There is,

⁵ Ho J *et al.* (2012). Conducted electrical weapon incapacitation during a goal-directed task as a function of probe spread. *Forensic Sci Med Pathol* **8**:358-366 (<https://doi.org/10.1007/s12024-012-9346-x>)

⁶ Despa F *et al.* (2009). Electromuscular incapacitation results from stimulation of spinal reflexes. *Bioelectromagnetics* **30**:411-421. (<https://doi.org/10.1002/bem.20489>)

⁷ Chiles BD *et al.* (2018). Electrical weapon charge delivery with arcing. *Annu Int Conf IEEE Eng Med Biol Soc* (<https://doi.org/10.1109/EMBC.2018.8512753>)

⁸ Sheridan RD and Hepper AE (2022). An analysis of officer-reported TASER X2™ probe discharge effectiveness in the United Kingdom. *J Forensic Leg Med* **91** (<https://doi.org/10.1016/j.jflm.2022.102417>)

⁹ Kunz SN *et al.* (2024). Effectiveness of a new-generation CEW in human subjects with a goal-directed task. *Hum Factors Mech Eng Def Saf* **8**:1 (<https://doi.org/10.1007/s41314-024-00066-x>)

therefore, a capability gap with the new device in comparison with earlier devices.

14. In addition to the warning alert, the TASER 10™ provides the officer with a beeping alert that begins two seconds before the end of a discharge cycle. Another alert sounds when probes have been fired and the TASER 10™ detects that there is an electrical connection to the subject. Provided that an electrical connection is still maintained with the subject, the connection alert also sounds when the default five-second discharge period terminates and the officer extends the cycle. While the connection alert can be disabled, SACMILL understands that a decision has been made by the NPCC to implement the function and monitor its utility post-authorisation of the TASER 10™ system. SACMILL has been advised, however, that a behavioural change observed in the subject takes primacy over the connection alert, partly because the latter would still activate when the probe spread of two or more skin-embedded darts was too narrow to elicit NMI *[paragraph 8]*.
15. When used to fire probes, the operational range of earlier CEDs is governed in large part by the divergence characteristics of the cartridge housing the probe pairs coupled with the need to secure a probe pairing on the subject. In this way, the maximum engagement range of the TASER X2™ and the TASER 7™ (with the latter fitted with the 'Stand Off' cartridge variant) is limited to 7.6 m (25 ft), although the practical engagement range will be lower due to the increased dispersion of probes away from the point of aim. With its individually fired probes and user-determined probe spread, the TASER 10™ is not constrained in the same way, with the result that the manufacturer claims a maximum range of 13.7 m (45 ft).
16. To achieve its increased range, the TASER 10™ probe is ejected from its cartridge case using the pressure generated from the combustion of an electrically initiated primer, whereas the probes of earlier CEDs were launched under nitrogen pressure. The probes of the TASER 10™ are propelled with greater energy than the nitrogen-propelled probes of the earlier devices.
17. Similar to the TASER 7™ probe, the electrically conductive wire tethering the TASER 10™ probe to the discharge-generating circuitry located within the handle of the device, is wound inside the probe body from where it spools out after the probe is fired.
18. The probe itself has several components: the probe body (containing the tethering wire) at the front end of which is a dart. Surrounding the base of the dart is an impact absorber designed to disperse some energy in the event that the dart penetrates tissue to its full extent. The dart length is about 11 mm when the impact absorber is uncompressed, which is a similar length to the darts of the TASER X2™ and TASER 7™ probes.
19. Like the upper probe of the TASER 7™, the TASER 10™ has a green laser sighting aid to assist in probe placement.
20. The TASER 10™ is switched on ('armed') or off ('made safe') by way of a multi-function selector switch which also serves to activate the warning alert, re-energise probes that are in electrical

contact with the subject, initiate a function test or engage 'stealth mode' (where the sighting laser and flashlight are disabled and the information display at the rear of the device is dimmed¹⁰).

21. Like other CEDs manufactured by Axon, the TASER 10™ has a display at the rear that serves to indicate the status of the device. Whereas the earlier devices used yellow-on-black displays, the TASER 10™ display uses yellow, red, blue and green icons on a black background. Device status is also indicated by the rail sidelights of the TASER 10™ which run either side of the device's 'muzzle'. The sidelight is yellow when the device is armed and fitted with operational cartridges but will show red if the device detects an internal error.
22. The above characteristics of the TASER 10™, together with SACMILL's opinion on the medical implications of these and other aspects of the TASER 10™ system, are discussed below.

The electrical output of the TASER 10™

23. The electrical output of four TASER 10™ devices was measured in independent testing using the method prescribed by Axon.¹¹ This method is designed to confirm the pulse rate, pulse charge and peak voltage of the waveform when measured across a 600 Ω load. The independent testing confirmed that the electrical output parameters conformed to the manufacturer's specifications.
24. The electrical pulse generated by the TASER 10™ is a monophasic, approximately rectangular waveform. When measured across a 600 Ω resistive load, the peak current, pulse duration, pulse charge and pulse repetition frequency (PRF) were found to be broadly similar to the equivalent measures for the output waveforms of the TASER X2™ and TASER 7™.¹² In line with expectations, there was no brief, very high voltage component at the start of the pulse.
25. The above testing involved measurement across only one of the forty-five potential output paths of the TASER 10™. However, there is no reason to suggest that the output parameters of the other forty-four paths would be different given that these are set by the firmware and electronic circuitry within the handle of the TASER 10™.
26. The PRF of the single path, which is 21-23 pulses per second, would be equivalent to that delivered to a person with two skin-embedded TASER 10™ probes. If a person is subjected to three or more skin-embedded TASER 10™ probes, Axon state that the maximum PRF to which the person would be exposed is 43-45 pulses per second¹³, although a slightly higher maximum rate of 45.7 pulses

¹⁰ After firing of the first probe, the device reverts from 'stealth mode' to normal function.

¹¹ TASER 10 Axon Certified Specification Test Procedure, version 5.0 (dated 24/06/2024). Axon Enterprise, Inc.

¹² The pulse charge of the TASER 10™ between any two connecting probes is understood to vary according to the resistive load sensed by the device, with a higher load being met with a higher pulse charge. The 600 Ω load is said by Axon to be a typical load measured in human subjects and loads have been seen to vary from around 300 Ω to 1000 Ω (personal communication to Dstl from Axon, 09/01/2025). According to Axon, the pulse charge measured at loads of 300 Ω and 2000 Ω is 52 and 95 µC, respectively, while the pulse charge delivered into a 600 Ω load is 70 µC (footnote 13). The independent testing confirmed the pulse charge and other parameters for a 600 Ω load.

¹³ TASER 10 Energy Weapon Specifications (dated 28/02/2023). Axon Enterprise, Inc.

per second was measured in independent testing [paragraph 6(j)]. The maximum PRF is broadly the same as that delivered by the TASER 7™ when both cartridges have been deployed and both pairs of probes are in electrical contact with the subject.

27. It is further understood from the manufacturer that, irrespective of the number of skin-embedded TASER 10™ probes that are in contact with the subject, each discharge pulse is only ever formed by the energisation of one pair of probes, although the identity of the probes forming the pair may change during the course of the discharge cycle. This electrical behaviour has now been verified in independent testing [paragraph 6(j)].
28. While the independent electrical testing focused on the Axon-recommended single load resistance, it is understood that the firmware continuously monitors the electrical resistance between deployed probes and adjusts the pulse charge accordingly, with the pulse charge being increased when the discharge is delivered into higher resistances and reduced for lower resistances.^{9,13} While this electrical behaviour was observed in independent testing, it was not systematically explored [paragraph 6(j)].
29. As with the UK's implementation of the TASER X2™ and TASER 7™, the default discharge cycle of the TASER 10™ is limited to 5-seconds, irrespective of whether the trigger is activated momentarily or continuously depressed. In this way, the officer has to make an active decision to extend the default cycle length should this be required.
30. The 5-second discharge cycle of the TASER 10™ is initiated when the second probe is fired, irrespective of whether an electrical connection is made. If a third probe is then fired, the 5-second cycle restarts, again, irrespective of whether a connection is made. This pattern continues until the final probe is fired and the 5-second cycle from this final shot has run its course.¹⁴

Ballistic aspects of the TASER 10™ probe

31. The initial independent technical testing [paragraph 6(c)] examined the intrinsic accuracy of probes fired from a clamped TASER 10™ device and measured their kinetic energy and momentum. Intrinsic accuracy is distinct from practical accuracy, which was separately explored using probes fired from handheld devices (see [User handling trials](#)).
32. When fired over target distances ranging from 1.5 m (5 ft) to 7.6 m (25 ft), the probes were observed to be accurate, with the dispersion of the impact points increasing with range, as would be expected. When fired over target distances of 10.1 m (33 ft) up to the nominal maximum range of the TASER 10™ of 13.7 m (45 ft), the cluster of impact points progressively dropped away from the point of aim and their dispersion increased.
33. The kinetic energy and momentum of the probes over the entire tested range were higher than the equivalent values previously measured for the probes fired from the TASER X2™ and TASER 7™. For

¹⁴ Personal communication to Dstl from Axon Enterprise, Inc (dated 10/01/2025).

example, at a range of 7.6 m (the maximum range of the probes fired from the earlier devices), the kinetic energy and momentum of the TASER 10™ probe were, respectively, some 3.4-fold and 2.2-fold higher than the TASER 7™ probe. The difference was even greater when comparing the equivalent values for the TASER X2™ probe.

34. The 'hook-and-loop' training (HALT) probes, which are designed to be used for scenario-based training and which attach themselves to a specialised suit worn by a role actor, were observed ballistically to behave very similarly to the probes that would be used in an operational setting.
35. The ten probes of the TASER 10™ are housed in individual cartridges which are inserted into a magazine attached to the front of the device. Intra- and inter-magazine variations in probe accuracy were examined in a limited number of tests. There was no significant difference in the mean impact points or dispersion between magazines for either the operational or HALT magazines.
36. Although the nominal maximum range of the TASER 10™ is said to be 13.7 m (45 ft), the true maximum range is determined by the length of the electrically conducting tethering wires that connect the probes to the TASER 10™ handle. The independent testing demonstrated that TASER 10™ probes could travel about 15 m (49 ft), at which point probes started detaching from their tethering wires. By 15.35 m, all probes had detached from their wires.
37. Given the higher energy of the TASER 10™ probes, the independent technical testing examined the potential of fired probes to injure the skin or produce skull fracture. Both tests used physical models and probes were fired at close range to approximate worst-case conditions. The testing compared the injury potential of both TASER 10™ and TASER X2™ probes, owing to the considerable amount of operational data available for the latter.
38. The skin surrogate model, which is intended to simulate the particularly vulnerable skin of a young child, showed that the bodies of both TASER 10™ and TASER X2™ probes had the potential to perforate the 'skin' beyond the dart. In several instances, both the impact absorber and part of the TASER 10™ probe body perforated the 'skin' [paragraph 18]. The TASER X2™ probe body similarly perforated the 'skin' on several occasions.
39. A skull surrogate model was used to assess the risk of skull fracture from the impact of the TASER 10™ probes. Comparative testing was conducted by firing the TASER 10™ and TASER X2™ side by side. None of the probes fired from either the TASER 10™ or TASER X2™ led to fracturing. Pin prick lesions of the surrogate skull, suggestive of some degree of penetration, were commonly observed with both types of dart. However, full perforation of the surrogate skull (by a TASER 10™ dart) was seen only once, which constituted an incidence of 5% of the TASER 10™ probe firings conducted during this phase of testing. The probe of this perforated dart could not be removed by hand and the dart was seen to be distorted to the extent that it had hooked back upon itself. In another case, a fragment of the TASER 10™ dart became embedded in the surrogate skull and the dart was seen to be distorted.

40. It was noted that the darts of the TASER 10™ probes in the surrogate skull penetration tests had, in general, deformed to a greater extent than the TASER X2™ darts. The most plausible explanation for this is the greater mass and velocity of the TASER 10™ probe and the impact of the dart with the flat bone portion of the model, resulting in buckling of the probes in fourteen of the twenty firings undertaken. In some cases, this resulted in the dart of the TASER 10™ probe deforming into a hook shape. Should this occur in an operational setting, probe removal could lead to significant damage to tissue than would otherwise have occurred had the dart been undeformed.

Clothing penetration

41. Owing to the requirement for the darts of the TASER 10™ probes to embed in the skin [paragraph 13], the initial independent technical testing [paragraph 6(c)] incorporated an assessment of dart penetration through various types of clothing at probe firing distances of 2, 8 and 15 m. The darts of the probes were found to reliably go through the limited range of clothing types tested. The additional testing included more challenging combinations of clothing. SACMILL understands the results from this work have been delivered to the College of Policing to ensure any implications are included in the training.

Laser sighting system

42. With the exception of the first probe firing in 'stealth mode' [paragraph 20], the approximate impact point of fired probes is guided by a green laser targeting aid said to be zeroed at 10.1 m (33 ft), meaning that the points of impact of probes fired at this distance should cluster around the laser dot. However, as noted above [paragraph 32], at 10.1 m the cluster of impact points had already fallen below the point of aim. Despite this, at firing distances less than 10.1 m there was no indication of probes impacting above the point of aim to any meaningful extent. This was further investigated during the additional tests [paragraph 6(i)], to examine any artefacts from the initial testing that could conflict with the reported zeroing range (such as clamped firing during some of the testing that would affect the zeroing range). No such conflicts were found.
43. The independent technical testing confirmed that the output of the laser targeting system was consistent with the manufacturer's Class 3R labelling.

Mechanical sighting system

44. In common with other CEDs, the TASER 10™ has a conventional mechanical sighting system that can be used for the first probe shot in 'stealth mode' or when there is difficulty visualising the laser dot, as might occur in bright sunlight or in the event of the laser failing.

Sound pressure level

45. The TASER 10™ uses a black powder ignition system to propel the probes from the magazine. The sound pressure level was measured during firing of the device and compared with the TASER X2™. Both were found to be within the health and safety legislation for impulsive auditory exposures, although the sound pressure level measured for the TASER 10™ was 1.7-fold higher than that measured for the TASER X2™.

User handling trials

46. User handling trials were conducted by the College of Policing to gain an insight into how the TASER 10™ performed in the hands of potential future users. Comparisons with the performance of the TASER X2™ and TASER 7™ were made where possible.¹⁵
47. The trial participants comprised officers who had previous experience of using CEDs and others who had never before used a CED.
48. The practical accuracy of the TASER 10™ was assessed in a series of exercises involving the use of specially designed target boards or role actors with the College of Policing concluding that the TASER 10™ has sufficient practical accuracy to be fired at up to 10 m using the laser sight. The College further concluded that, in the hands of experienced CED users, the device remained accurate when using the mechanical sight to assist aiming at 10 m, but that greater accuracy at this distance was still achieved using the laser sight. The drop-off in probe trajectory at 10 m, seen in the intrinsic accuracy tests [paragraph 32], was not evident in the user handling trial.
49. Due to the degradation in probe accuracy and consistency with range [see paragraph 42], the College of Policing concluded that the optimum effective range (now referred to in training as the ‘maximum practical range’) of the TASER 10™ should be regarded as 10 m, accepting that the device may still be accurate out to its nominal maximum range of 13.7 m.
50. SACMILL has reviewed the user handling trial report [paragraph 6(d)] and the Committee is broadly in agreement with the College’s interpretation of the findings around the practical accuracy of the TASER 10™. SACMILL also sought confirmation that none of the results from the additional testing would change the College’s findings.
51. One exercise explored the potential value of the extended range of the TASER 10™. The exercise involved a ‘threatening’ role actor approaching the officer from an initial distance of 15 m (49 ft). The officer was asked to fire probes when they thought the role actor was at a suitable firing distance. The firing distances ranged from 2.9 to 11.3 m with a median of 8.3 m. Most of the shots taken were beyond the practical firing range of either the TASER X2™ or the TASER 7™ (when fitted with the ‘Stand Off’ cartridge variant).

¹⁵ The increased range of the TASER 10™ meant that some exercises were unique to the new device.

52. In addition to the probe firing evaluations, the user handling trial solicited the trial participants' opinions on various aspects of the TASER 10™ device. Where considered relevant, these opinions have informed the [Medical implications of the TASER™ 10 system](#).

Training and assessment documentation from the College of Policing

53. The training modules available for SACMILL review were at an advanced stage of development. There were notable changes from previous versions in that the various firing stances necessary with legacy devices to increase stand-off are not necessary with the TASER 10™ because each probe is individually aimed. Also, there is no need with the TASER 10™ to tilt the device when firing at subjects in challenging postures, as may have been necessary with previous CEDs. These are both seen as reducing the system complexity.
54. Although the nominal maximum range of the TASER 10™ probe is 13.7 m (45 ft), the College of Policing have chosen a shorter range over which the use of the device was seen to be acceptable. This 'maximum practical range' is understood to be partly to optimise the accuracy of probe placement and partly to mitigate issues with egocentric range estimation (the tendency for people to underestimate the distance to objects in their visual field).¹⁶ The reduced range also lessens the risk of firing at an individual who is beyond the maximum range of the probe wire, where there is a risk that probes will detach from their wires to become a free-flying hazard [*paragraph 36*]. The reduced range also increases the probability of firing at an individual who is within the maximum length of the probe's tethering wire.
55. The training covers the various features of the TASER 10™, however, the reliance on the single multi-function selector switch will require significant familiarisation to ensure the functions are correctly selected during dynamic operational incidents. The training is intended to ensure the use of this switch is suitably understood by users.
56. A virtual reality system, which is designed to complement conventional training, has been noted (and observed by SACMILL). In particular, it was possible to note how training could be augmented, especially through the creation of scenarios that would be difficult and expensive to recreate in the classroom or in range-based training.
57. Formative learning is similar to previous CED training and includes classroom instruction, practical exercises, procedural drills and live fire, alongside scenario-based training and summative assessments of competence, written knowledge checks and a qualification shoot. Summative assessments have been adapted to reflect the new features of the TASER 10™; the qualification shoot includes multiple probe placements at various ranges and tests the trainees' ability to operate the multi-function selector switch correctly. The formative training shoots and scenarios

¹⁶ Kelly JW *et al.* (2004). Judgments of exocentric direction in large-scale space. *Perception* **33**:443-454. (<https://doi.org/10.1068/p5218>)

have been adapted to support increased accuracy of individual probe placement, the requirement for skin penetration and use of the connection alert.

NPCC and College of Policing implementation programme for the TASER 10™ system

58. SACMILL has seen the joint NPCC and College of Policing TASER 10™ Implementation Programme [paragraph 6(g)]. This provides advice to forces planning to introduce the TASER 10™ and highlights some of the concerns that have driven the introduction of the new device, such as the reliance on earlier devices that are no longer fully supported by the manufacturer.
59. The TASER 10™ Implementation Programme also raises concerns that need to be considered by forces, such as:
- (a) How forces will use and manage devices when they have a CED capability consisting of multiple models. To mitigate this, officers already trained and accredited to use an earlier model of CED will not be permitted to revert back to that device once trained and accredited to deploy operationally with the TASER 10™, although exceptions apply [paragraph 83].
 - (b) The implementation programme states the need for good management of the system, especially as the firmware updates are managed through the battery system and docking with the data management system. Good management will also be required to ensure preventative maintenance for cleaning and the replacement of certain parts of the handle. Forces will need to ensure that this management is in place. Forces will also need to ensure that all training staff have undergone the relevant training before they in turn can provide training to others.
60. SACMILL has also seen a draft fact sheet produced by the NPCC to accompany the release of the system [paragraph 6(h)]. This fact sheet anticipates a perceived benefit of the TASER 10™ in that it may provide additional de-escalation options, rather than increasing the number of probe deployments. This aspiration, which remains to be evidenced in operational practice, should also be viewed in the context of the de-escalation potential of the new audio-visual warning display of the TASER 10™ [paragraph 13] versus the ‘crackling’ arc display of earlier devices.

Medical implications of the TASER 10™ system

61. SACMILL’s opinion on the medical implications of the TASER 10™ system has been informed by the various elements of the system outlined in the foregoing paragraphs. Additionally, SACMILL considered several Axon-sponsored studies on the TASER 10™. Owing to the new design features of the TASER 10™ compared with previous CEDs, SACMILL specifically requested human factors advice to assist its understanding of the system under consideration.¹⁷

¹⁷ Human factors input was provided by Dstl and formed part of the report supplied to SACMILL [paragraph 6(a)].

Electrical output of the TASER 10™

62. The output pulse waveform of the TASER 10™ [paragraphs 23-30] lacks the brief, very high voltage (ca. 50 kV) component that enables the discharge from earlier devices to arc across a small air gap. This means that an arc warning display can no longer be presented to the subject. The effectiveness of the arc display of the earlier devices as an aid to de-escalation is unclear and, therefore, the implications of its absence in the TASER 10™ cannot be assessed at the moment. However, this is something that can be evaluated in operational reporting should the TASER 10™ be authorised for use.
63. The absence of the very high voltage component also means that there is no longer a drive-stun option nor the possibility to follow-up probe firing with direct contact with the subject with the electrodes at the front of earlier devices. The loss of both of these options may detract from the operational effectiveness of the TASER 10™ compared with earlier devices. Here again, analysis of operational reporting may help to provide insights into any operational shortfalls in this regard.
64. A further consequence of the loss of the very high voltage component of the waveform is that the output pulse can no longer arc across a small air gap and traverse clothing to make an electrical connection with the subject. This means that the darts of the TASER 10™ probe must embed in skin to induce NMI. The ability of one or both probes of the earlier devices to overcome clothing to some degree has been shown to contribute to the subdual effectiveness of probe discharge, although the highest subdual rates were reported when the darts of both probes embedded in the skin.⁸ Provided that at least two skin-embedded darts can be reliably obtained with the TASER 10™ probe, the loss of waveform arcing is not likely to adversely influence the subdual effectiveness of the proposed new system. However, this opinion should be tested by close monitoring of operational outcomes.
65. The pulse repetition frequency and pulse charge characteristics of the TASER 10™ are broadly similar to those generated by the earlier twin-cartridge devices [paragraphs 23-26]. The waveform shape of the TASER 10™ is approximately rectangular and much less complex than the waveform shapes delivered from earlier devices. Animal studies indicate that rectangular waveforms are at least as effective as complex waveforms at stimulating muscle contractions.¹⁸
66. SACMILL noted the manufacturer's original claims relating to pulse discharge produced by the TASER 10™, however the Committee requested independent confirmatory testing to clarify the discharge situation in instances where more than two probes are in electrical contact with a subject since this is such a fundamental feature relating to safety. The independent testing has reported [paragraph 6(j)] and has shown that the device forms each discharge pulse through the energisation of a single pair of probes. The combination of probe-pairings contributing to each pulse may change during the course of the discharge cycle, but only one pair is active during each

¹⁸ Comeaux JA *et al.* (2011). Muscle contraction during electro-muscular incapacitation: a comparison between square-wave pulses and the TASER X26 electronic control device. *J Forensic Sci* **56**:S95-S100. (<https://doi.org/10.1111/j.1556-4029.2010.01580.x>)

pulse. Pulses were found to discharge at a rate of up to 45.7 pulses per second in this testing [see paragraph 26].

67. The physiological effects of TASER 10™ discharge in human volunteers have been evaluated in an Axon-sponsored study.¹⁹ The discharge was administered for eight seconds through multiple skin-embedded probes positioned in various body locations. A number of physiological and biochemical parameters were monitored before and during exposure, one-hour post-exposure and on the following day. The authors reported that the findings with the TASER 10™ were consistent with their previous findings on the physiological effects of other Axon CEDs. However, five of the twenty-two subjects exposed to discharge via at least two probes located on the back, became apnoeic (stopped breathing) throughout the duration of the eight second exposure. Although the authors had not previously reported apnoea in their earlier CED studies, cessation of ventilatory effort induced by TASER X26™ discharge to the back has been reported by another group.²⁰ In addition to apnoea, the Axon-sponsored study reported syncope (fainting) in two participants at the end of the eight-second exposure. Syncope was observed in one participant in an Axon-sponsored study of the human effects of the TASER 7™.²¹ This latter event, which also occurred at the end of the discharge period, prompted SACMILL to cite it in the Committee's medical statement on the TASER 7™, where it was described as a novel injury mechanism which could increase the risk of injury from falls.⁴ SACMILL reinforces the need to terminate exposure to discharge once the desired effect has been achieved; this is due to the potential risks from prolonged exposure, including respiratory arrest. Discharge-induced apnoea and syncope are a concern. The mechanisms for each remain unclear but apnoea could be due to tetany of the accessory muscles of respiration. Syncope can be due to a number of reasons, amongst which are transient cardiac dysrhythmia and even the pain induced by the CED discharge.²² Should reporting identify a significant rate of apnoea or syncope in operational deployment, SACMILL must be made aware so that the Committee may reconsider these potentially dangerous effects.
68. The ability of the TASER 10™ discharge to induce NMI has been reported in a second Axon-sponsored human volunteer study, where the effectiveness of the electrical discharge delivered from the device was compared with that delivered from the TASER 7™.²³ The authors reported that the effectiveness of both devices was similar. While SACMILL finds this reassuring, it remains important that the subdual effectiveness of the TASER 10™ under operational conditions is monitored in the event that the new CED system is authorised for use in the UK.

¹⁹ Dawes DM *et al.* (2024). Physiologic changes with an exposure to a new concept conducted electrical weapon (T10™) in human volunteers. *Hum Factors Mech Eng Def Saf* 8:3 (<https://doi.org/10.1007/s41314-024-00067-w>)

²⁰ VanMeenen KM *et al.* (2013). Respiratory and cardiovascular response during electronic control device exposure in law enforcement trainees. *Front Physiol* 4:78. (<https://doi.org/10.3389/fphys.2013.00078>)

²¹ Ho JD *et al.* (2020). The physiologic effects of a new generation conducted electrical weapon on human volunteers at rest. *Forensic Sci Med Pathol* 16:406-414. (<https://doi.org/10.1007/s12024-020-00249-w>)

²² *Syncope (Fainting)*. American Heart Association (<https://www.heart.org/en/health-topics/arrhythmia/symptoms-diagnosis--monitoring-of-arrhythmia/syncope-fainting#:~:text=Syncope%20is%20a%20symptom%20that,restores%20blood%20flow%20and%20consciousness.>)

²³ Kunz SN *et al.* (2024). Effectiveness of a new-generation CEW in human subjects with a goal-directed task. *Hum Factors Mech Eng Def Saf* 8:1 (<https://doi.org/10.1007/s41314-024-00066-x>)

69. The cardiac effects of TASER 10™ discharge in an anaesthetised pig model have been reported in a third Axon-sponsored study in which the discharge was applied with the darts of the TASER 10™ probes fully inserted into the chest.²⁴ Using blood pressure drop as a surrogate indicator of cardiac capture or malignant arrhythmia, the authors reported arterial blood pressure decreases consistent with cardiac capture but no induction of ventricular fibrillation. The risk of CED-induced cardiac capture, together with its potential consequences in the human population, was long-ago articulated by SACMILL's predecessor²⁵ and remains a concern for SACMILL.
70. Although lacking the initial very high voltage component, the waveform peak voltage still approaches 1 kV.²³ While this means that the TASER 10™ waveform is unable to arc across a meaningful air gap, it does not mean that smaller-scale electrical sparking will be absent. SACMILL understands that the dangers associated with discharging a CED in an explosive environment, or where the subject is covered in a flammable liquid, are already included in the training for existing CEDs; this should be retained for the TASER 10™ unless the ignition risk posed by the discharge of the new device has been deemed in independent testing to be manageably low.

Ballistic considerations and injury potential of the TASER 10™ probes

71. In an effort to increase the likelihood of obtaining skin-embedded darts, as well as increasing the maximum usable range of the TASER 10™, probes fired from the device have greater kinetic energy and momentum than probes fired from the earlier devices [paragraph 33].
72. SACMILL notes a fourth Axon-funded pilot study in which TASER 10™ probes were fired at the bare abdomen, thigh, buttock and back of 20 adult human volunteers.²⁶ The study was prompted by the high velocity of the probe and the need to exclude the possibility of what the authors described as 'tissue overpenetration'. When the probes were fired at a distance of 60 cm (24 inches), the authors reported signs of epithelial abrasion at the targeted sites but no evidence of penetration of the probe body through the skin. The authors acknowledged, however, that two limitations of the study were the small sample size and that the targeted regions were confined to Axon's preferred target areas.²⁷ Therefore, impacts to other sites, such as the skull, neck or groin were not evaluated.
73. SACMILL has concerns that the higher kinetic energy and momentum of the TASER 10™ probe has the potential to produce a greater degree of blunt impact injury than seen with probes fired from earlier devices, despite the presence of an impact absorbing component surrounding the base of

²⁴ Dawes DM *et al.* (2025). Cardiac safety using a swine surrogate model for a new concept conducted electrical weapon—the TASER® T10. *J Transform Tech Sustain Dev* 9:1 (<https://doi.org/10.1007/s41314-025-00068-3>)

²⁵ Statement on the Medical Implications of Use of the Taser X26 and M26 Less-Lethal Systems on Children and Vulnerable Adults. DOMILL statement dated January 2012. (https://assets.publishing.service.gov.uk/media/5a7f3224ed915d74e33f4ebc/DOMILL14_20120127_TASER06.2.pdf)

²⁶ Ho JD *et al.* (2023). Safety profile of new TASER Conducted Electrical Weapon darts. Presented at the 11th Symposium on Non-Lethal Weapons, Brussels, 22nd-25th May, 2023.

²⁷ TASER 10 Energy Weapon User Manual, Rev C, November 2024. (https://my.axon.com/s/taser-10?language=en_US)

the dart of the TASER 10™ probe [paragraph 18].

74. Although the exposed dart length of the TASER 10™ probe is similar to that of previous devices, the higher kinetic energy and momentum will translate into a higher risk of penetrating injuries to deeper-lying organs and tissues due to the elastic deformation of the body wall at the dart's point of entry as well as the deformation of the impact absorbing component situated at the base of the dart [paragraph 75].
75. While the dart length of the TASER 10™ probe is 11 mm when the impact absorber is in its uncompressed state [paragraph 18], the independent testing demonstrated that the length of the dart reaches 15 mm when the impact absorber is compressed to its fullest extent, a situation that might conceivably be approached were the dart to fully insert into a structure such as a rib, the skull, other bony tissue or some areas of soft tissue. That the darts of the probes of existing CEDs are able to pierce through the skull has been documented in several case reports in the medical literature.²⁸ SACMILL therefore recommends and reiterates that the head, face and neck should never be intentionally targeted unless under wholly exceptional circumstances.
76. SACMILL notes the possible complications arising if the dart of the TASER 10™ probe enters bone [paragraph 39], where it has the potential to distort in such a way that might complicate removal (including increased risk of bleeding, infection and retention in bone). Dart fragment retention leads to an increased risk of development of osteomyelitis (bone infection). This highlights the need for CED users to have some grasp of the possible complications and for appropriately trained health care professionals to examine people subjected to CED discharge, a position endorsed by the NPCC Less Lethal Weapons National Lead.²⁹
77. Probes should be removed only by personnel who have been trained in the correct removal techniques. Training for the users of TASER 10™ should emphasise the increased risk of medical complications compared with previous devices.

Training and assessment documentation

78. The College of Policing's adoption in training of guidance on the 'maximum practical range' of the TASER 10™ [paragraph 54] is seen as a pragmatic approach intended to reduce the risk of free-flying probes and maximise the accuracy of the device (and minimise the risk of strikes to vulnerable parts of the body). SACMILL also notes improvements to the training programme to assist with range estimation [paragraph 90], but retains some concern over how well TASER 10™ officers will be able to apply this guidance due to errors in range estimation.¹⁶
79. The increased complexity of the device, in respect of the Central Information Display, multi-function selector switch and audible alerts, are covered in training. However, whether this added

²⁸ Shetty A *et al.* (2024). TASER dart causes penetrating intracranial injury. *BMJ Case Rep* 17:e261108. (<https://doi.org/10.1136/bcr-2024-261108>)

²⁹ NPCC National Circular 05LL'2021, dated 26/04/2021. (<https://fflm.ac.uk/wp-content/uploads/2021/05/05LL2021-Healthcare-of-detainees-in-custody-subjected-to-CED.pdf>)

complexity, compared with earlier CEDs, would translate into an increased incidence of user error in operational service is yet to be understood.

80. SACMILL notes the availability of a virtual reality (VR) environment which may reinforce classroom or range-based training.
81. Training is supplemented with videos of use in a benign (demonstration) environment. It is noted that more than two probes were required in this demonstration to achieve NMI, which will reinforce the need to be prepared to follow-up with additional probe firings if the first two do not result in the required behavioural change indicative of NMI.

NPCC and College of Policing implementation programme for the TASER 10™ system

82. The review of the system by SACMILL has had to assume that the comprehensive management system specified by the NPCC and College of Policing in their implementation programme for the TASER 10™ is introduced – failure to correctly manage the system and the authorisation of officers will increase risk both to the public and to officers.
83. The current version of the NPCC and College of Policing implementation programme seen by SACMILL [*paragraph 6(g)*], states that officers will only be authorised in one version of CED for operational use. Whilst some specialist officers, for example CED instructors may be familiar with more than one design of CED, they will only be authorised to be issued with one model of CED. The implementation plan states that one of the key principles for the authorisation of use of the TASER 10™ is that “procedures must be in place that ensures users are only issued with the device for which they are currently authorised”.
84. SACMILL welcomes the NPCC’s aspiration, articulated in their fact sheet [*paragraph 6(h)*], that the TASER 10™ promises to provide additional de-escalation options, especially in light of the recommendation from the Independent Office for Police Conduct that a sufficient emphasis is placed in police conflict management guidelines on communication and de-escalation techniques.³⁰ How the aspiration for increased de-escalation opportunities will be realised and evaluated for the TASER 10™ has yet to be set out.

Human factors considerations

85. The human factors elements considered by SACMILL covered trust in the system, individual factors and situation factors.
86. The College of Policing’s user handling trials [*paragraph 6(d)*] were undertaken in three phases, with the majority of the work being conducted during the first phase. However, faults were identified during the first phase. After liaising with the manufacturer, the College conducted a

³⁰ Independent Office for Police Conduct (IOPC). Review of IOPC cases involving the use of TASER 2015-2020. 01/08/21 (<https://www.policeconduct.gov.uk/publications/review-iopc-cases-involving-use-taser-2015-2020>; accessed 28/01/2025)

smaller-scale second trial. This revealed continued reliability issues which necessitated a small third trial, again after liaising with the manufacturer. Throughout the course of the three trials, the frequency of faults decreased but did not disappear entirely. Faults were also seen in the technical testing [paragraph 6(c)] albeit at a reduced rate, and some types of fault seen in the user handling trials were not manifested in the technical testing.

87. SACMILL has been advised that the progressive reduction in the incidence of faults over the course of the trials was seemingly as a result of the manufacturer's intervention in the form of revisions to the device handle and updates to the device's firmware. SACMILL endorses the College's view that, in the event that the TASER 10™ system is authorised for use, reliability should continue to be closely monitored and improved.
88. In its reporting, the College cautioned that the findings in the user handling trials should be interpreted in the context of the specific handle and firmware revision used in each trial phase. Although the fewest faults were detected in the final phase of the trials, the College cautioned that the findings may not necessarily be valid for future handle or firmware revisions. SACMILL concurs with the College's cautionary note.
89. In the user handling trials, the practical accuracy of probes fired from the TASER 10™ using the laser sight was found to be good up to a target distance of 10 m [paragraph 48]. The practical accuracy at this distance using the mechanical sight, however, was not as good. Beyond 10 m, the practical accuracy degraded, and the dispersion increased, resulting in an increased likelihood of missing the intended target. This has resulted in the College of Policing using the terminology of a 'maximum practical range'.
90. It is understood that humans have a propensity to underestimate an object's distance.¹⁶ In the present context, this means that TASER 10™ officers may fire probes at subjects who are further away than the officers perceive, resulting in a reduced targeting accuracy. SACMILL acknowledges that there is an intention that range estimation will be taught in TASER 10™ training using the length of a familiar object, such as a car, as a reference.
91. As well as the practical accuracy of the TASER 10™, trust in the system is dependent upon the functionality of the device and the ability to use the device with ease and without error. The multi-function selector switch [paragraph 20] serves to control more functions on the TASER 10™ than the equivalent switch on earlier devices and, therefore, involves greater complexity. While 70% of the participants in the user handling trials were of the view that the TASER 10™ switch was easy to operate and that it was easy to perform a warning alert [paragraph 13], that left a significant proportion who disagreed with both of these assertions.
92. The gist of the negative comments received from a third of the user handling trial participants regarding the TASER 10™ warning alert, was that it was difficult to apply and maintain the upward pressure on the switch needed to activate the alert. Notably, after probes have been deployed and the subject has been incapacitated, an identical manipulation of the selector switch is required to

extend the discharge cycle, should this be necessary.

93. A number of officers participating in the user handling trial questioned the deterrence value of the TASER 10™ warning alert *[paragraph 13]*. SACMILL is of the view that this is something that should be evaluated further in operational use in the event that the system is authorised for use. Furthermore, eight of twenty-seven trial participants disagreed with the assertion that the multi-function selector switch *[paragraph 20]* was easy to use or that it was easy to perform a warning display. How this might translate in a complex and dynamic operational environment is yet to be determined and is something that should be evaluated during operational use.
94. The individually fired probes of the TASER 10™ appear to confer a reduction in system complexity. The two most recent predecessors to the TASER 10™ (the TASER X2™ and TASER 7™) project two laser dots, one for the upper probe and one for the lower probe, with the separation of the laser dots projected onto the subject being determined by the cartridge divergence and the distance of the subject. This effectively creates a minimum distance at which the probes of the older devices may be fired (whilst maintaining an adequate probe spread) and a maximum distance (where the spread of the probes will be too big, risking probe strikes to the head and neck or a probe miss). With the individually fired probes of the TASER 10™ and probe spread no longer constrained by the cartridge divergence and the subject's distance, there is no lower limit to the firing distance and the upper limit is the maximum distance over which the TASER 10™ can be fired whilst maintaining an acceptable degree of accuracy. The concept of use of the TASER 10™ may also be helpful with quickly moving subjects, which TASER X2™ officers reported to be the case in nearly half of incidents in which probes of this earlier device were fired.⁸
95. The individually fired probes of the TASER 10™ may also reduce complexity where there is a need to incapacitate a subject in a challenging posture (for example, sitting or lying down) or who is partially obscured (for example, by furniture or a door) *[paragraph 53]*.
96. The presence on the TASER 10™ of a single Class 3R green (510 nm) laser sighting aid is welcomed by SACMILL. The TASER 7™ has an identical green laser to assist in the targeting of its upper probe. Dstl has informed SACMILL that users of the TASER 7™ have commented on the high visibility of the green laser, even in bright sunlight, consistent with the laser's wavelength being close to the peak sensitivity of the wavelength response of the human eye. The risk of injury from ocular exposure to Class 3R lasers is relatively low for short and unintentional exposures.³¹ The high visibility of the green laser may lessen complexity for the user.
97. A reduction in complexity may also be anticipated in circumstances involving the use of the TASER 10™ on aggressive animals. The individually fired probes of the TASER 10™ may facilitate obtaining a successful probe pairing on the animal without the constraints imposed by a fixed cartridge divergence. The TASER 10™ also permits a greater stand-off distance due to the increased range of its probes.

³¹ <https://www.gov.uk/government/publications/laser-radiation-safety-advice/laser-radiation-safety-advice>

98. The need for the officer to maintain situation awareness over increased distance (due to the range of the device), whilst potentially firing a greater number of probes, introduces additional complexity in terms of ensuring that bystanders are not unintentionally targeted. Additionally, that the probes of the TASER 10™ can detach from their tethering wires [paragraph 36] is a phenomenon first noted as part of SACMILL's assessment of the TASER 7™, where it was identified as a new injury risk for officers and bystanders located down-range of the subject in the event of probes missing.⁴ To address this, backdrop awareness was introduced into TASER 7™ training and a similar approach appears to have been adopted for the TASER 10™.
99. Whereas earlier devices made by Axon used monochromatic (yellow-on-black) displays, the TASER 10™ has a multicoloured display [paragraph 21]. This may have implications for officers with certain types and severities of colour vision deficiency.
100. The strobing effect that is part of the TASER 10™ warning alert [paragraph 13] alternates the device's torch intensity between 1000 and 210 lumen at a frequency of 20 Hz. While this strobing frequency lies within the range able to trigger seizures among those with photosensitive epilepsy, the incidence of this condition in the UK is reportedly less than 1 in 3,000.³²
101. SACMILL has been made aware of some limited anecdotal evidence from two international law enforcement bodies who have adopted the TASER 10™. This indicates an average of 3.3 probe deployments (in forty-four incidents) with seven or more probes being fired in five incidents. The average is broadly in line with Dstl's analysis of UK TASER X2™ data [paragraph 6(a)], which estimated that an average of four to six probe firings would be needed to achieve a 95% or more probability of achieving two skin-embedded darts.³³ Fewer probe firings might be anticipated with the TASER 10™ due to the higher energy of the probes and the accompanying expectation that these are more likely to overcome clothing to embed in the skin.

Conclusions

102. The TASER 10™ is a novel CED concept in which up to ten probes are individually fired. A minimum of two of these probes must embed in the skin of the subject for NMI to be induced.
103. A number of features of the TASER 10™ were considered by SACMILL to decrease system complexity compared with previous devices. These include the high visibility single green laser sighting aid and officers themselves determining the probe spread necessary for NMI rather than it being determined by the cartridge type being used and the subject's distance, as was the case with earlier CEDs. With the TASER 10™, there is also no need to tilt the device when attempting to

³² Epilepsy Action: <https://www.epilepsy.org.uk/info/seizure-triggers/photosensitive-epilepsy#:~:text=People%20with%20photosensitive%20epilepsy%20are,most%20likely%20to%20trigger%20seizures> (accessed 28/01/2025).

³³ The analysis was based on the officer-reported skin (versus clothing) penetration of the dart of the upper probe of the TASER X2™. The lower and upper estimates of the number of probe firings required relate to summer and winter months, respectively.

obtain a probe pairing on subjects in challenging postures or when partially obscured by objects.

104. Adding to system complexity, however, is the requirement to obtain at least two skin-embedded darts, often in fast-moving incidents, and the multi-function switch which is used to select more device functions than was the case on earlier CEDs. The amount of data conveyed to the officer in the information display located at the rear of the device also adds to system complexity as do the multiple types of audible alerts emitted from the TASER 10™ speaker port that indicate device status. For all CEDs, probes that miss the subject have the potential to present a hazard to police and bystanders located down-range of the subject. Additionally, for the TASER 7™, and now for the TASER 10™, probes that miss the subject can detach from their tethering wire to become a free-flying hazard. The higher kinetic energy, momentum and range of the TASER 10™ probe means that officers will need to be particularly aware of the backdrop hazard when firing the device. This also brings added system complexity.
105. From the subject's perspective, the electrical effects of the TASER 10™ discharge will likely be broadly similar to previous twin cartridge devices when one or two cartridges have been deployed. SACMILL, therefore, does not anticipate any additional electrical risks to emerge with the TASER 10™ other than those already set out by the Committee in this and its other CED medical statements.^{3,4} These risks include, but are limited to, excessive physiological stress from prolonged exposure to discharge (especially if apnoea occurs during the discharge period) and cardiac capture from a direct action of the discharge in circumstances where a dart has penetrated the frontal chest and the tip of the dart is sufficiently close to heart tissue. Head injury as a result of discharge-induced falls remains a concern for SACMILL. Although the neuromuscular effects of the discharge are the most likely mechanism for inducing falls, a second mechanism is syncope, which was reported in two Axon-sponsored studies in a small number of participants at the end of the discharge period [*paragraph 67*].
106. The kinetic energy and momentum of the TASER 10™ probes are higher than with previous devices. This is likely to raise the severity of skin wounds produced by probes fired from the new device and the dart of the probe has the potential to produce deeper penetrating injuries. Moreover, while a proportion of the NMI responses could be achieved with earlier devices when one or both probes were in clothing⁸, to achieve the same NMI outcome with the TASER 10™ the darts of the probes must embed in the skin. In tandem with the increased energy of the TASER 10™ probe, this means that the risk of significant penetrating injury will be higher than before. The skull, pleural cavity, abdominal organs and major blood vessels of the groin, neck and axillae may be at increased risk of direct injury from the TASER™ 10 probes and this may lead to an increased incidence of potentially life-threatening injuries.
107. The less mature anatomy of younger children may make them more susceptible to discharge and probe injuries.²⁵ Such susceptibilities include a thinner body wall in younger children which puts underlying tissues and organs (including the heart) at greater risk from skin-perforating TASER 10™ probes.²⁵ Furthermore, a thinner skull in younger children may mean that skull-perforating darts have greater potential to injure the brain and its surrounding tissues than they would in the mature adult.²⁵ Discharge-induced falls may have more serious consequences for younger children

due to a higher risk of bone fracture arising from bone immaturity during the adolescent growth period.²⁵ The latter, however, would be mitigated to an extent by the better energy-absorbing properties of adolescent bone.^{34,35} The smaller height of younger children may carry its own inherent risk as the vulnerable head and neck regions may be closer to the point of aim should probes be fired towards the upper body.

108. SACMILL understands that it is for individual forces to conduct an Equality Impact Assessment (EIA) in advance of authorising the TASER 10™ system for operational use. The EIA will allow Chief Officers to satisfy themselves that all the available evidence has been considered relating to any impact on groups with protected characteristics and to provide mitigation where appropriate.
109. SACMILL understands that CED training already includes advice to avoid firing probes at vulnerable areas, highlighting the eyes, head, neck and groin. SACMILL notes, however, that other areas are also vulnerable. These include, but are not limited to, the underarm (an area rich in neurovascular structures), any peripheral nerve or major blood vessel susceptible to injury from the probe dart and the pleura (with isolated case reports of dart-induced pneumothorax). The training also recognises the potential for the induction of cardiac arrhythmia should the dart of the probe fully embed in a region of the frontal chest and come to rest close to the heart [*paragraph 69*]. Officers, health care professionals and custody staff need to be aware of the greater potential for injury from TASER 10™ probes and be in a position to react expediently to any particular health concerns observed with subjects on whom probes have been deployed.
110. SACMILL takes a particular interest in the potentially increased opportunities for de-escalation that the longer engagement range of the TASER 10™ promises to provide [*paragraph 84*]. If this perceived benefit materialises in operations, SACMILL would anticipate seeing it reflected in a decrease in the proportion of incidents in which probes are discharged compared with shorter range CEDs. If so, this should be manifested in the annual use-of-force statistics published by the Home Office.
111. SACMILL requested additional work to go forward into independent technical testing. This work has fully reported and is now reflected in this updated medical statement
112. Finally, SACMILL advises that people in custody who have been exposed to CED discharge should be reviewed by a nurse, paramedic or doctor who has received specific training in the range of medical complications that might present.^{29,36}

³⁴ Ouyang J *et al.* (2003). Biomechanical character of extremity long bones in children. *Chin J Clin Anal* **21**:620–623

³⁵ Forman JL *et al.* (2012). Fracture tolerance related to skeletal development and aging throughout life: 3-point bending of human femurs. 2012 Conference proceedings of the IRCBI (International Research Council on the Biomechanics of Injury).

³⁶ Faculty of Forensic and Legal Medicine CED Hub: <https://fflm.ac.uk/cedhub/> (accessed 28/02/2025)

Recommendations

113. Recommendation 1

The greater energy and momentum of the TASER 10™ probes mean that injuries to sensitive areas, such as the eyes, head, neck, groin or axillae, may be more serious than previously encountered. This makes it all the more important that these areas are avoided unless considered reasonable and proportionate in the circumstances. Given the greater accuracy of the TASER 10™ compared with earlier devices, SACMILL recommends that an increased emphasis on targeting the thighs may be warranted, particularly for shots aimed towards the front of the subject.

114. Recommendation 2

Given the findings around device reliability in the user handling trials, SACMILL is of the view that measures should be put in place by the NPCC to closely monitor the on-going performance of the TASER 10™ and to react in a timely way should problems emerge. SACMILL wishes to be kept informed on this aspect as operational experience of the TASER 10™ system is acquired.

115. Recommendation 3

As part of its TASER 10™ training, the College of Policing should consider the possibility that some subjects on whom probe discharge is used, could become compliant after the first probe has been fired and that officers should be alert to this possibility and react accordingly.

116. Recommendation 4

Given the extended engagement range of the TASER 10™, set against the background of distance underestimation, the College of Policing should consider monitoring the effectiveness of the guidance on range estimation in training. Such monitoring may assist in minimising the possibility that probes are fired at subjects located at a distance where accuracy is degraded or who are beyond the range at which probes may become detached from their tethering wires and become a free-flying hazard.

117. Recommendation 5

The College of Policing should seek advice from a colour vision specialist on any difficulties officers with various types and severities of colour vision deficiency might encounter with the multicoloured Central Information Display of the TASER 10™.

118. Recommendation 6

The NPCC fact sheet for the TASER 10™ mentions increased opportunities for de-escalation due the greater engagement range of the TASER 10™. Whether these opportunities arise in practice should be the subject of in-service review having been captured in the operational reporting. This should be reported to SACMILL at the same time as other routine reporting.

119. Recommendation 7

The increased complexity of the multi-function selector switch raises the possibility that this will be reflected in increased errors in function selection arising in operational service. The user experience with the device should, therefore, be monitored in service.

120. Recommendation 8

Consideration should be given by adopting forces to the use of virtual reality systems to augment classroom and range-based training.

121. Recommendation 9

Given the reduced arcing potential of the TASER 10™ discharge, the risk of ignition of explosive vapours or flammable liquids of certain volatile organic compounds, such as propane or petrol, should be independently tested before changing the current training advice around the use of CEDs in the presence of these chemicals.

122. Recommendation 10

In independent testing, the sound pressure level generated by the TASER 10™ during probe firing was within the health and safety legislation for impulsive auditory exposures. Nevertheless, compliance with legislation should be confirmed in training establishments, to ensure that staff and trainees are not exposed to excessive cumulative noise through repeated exposures.

123. Recommendation 11

Should the TASER 10™ system be authorised by the Home Office, it should be mandatory that its operational performance is monitored to obtain reassurance that the system performs in the manner anticipated. This should continue for a minimum of twelve months and SACMILL must be informed immediately of any adverse, or potentially adverse, medical outcomes that may have a bearing on the advice set out in this statement. SACMILL anticipates formal reporting on the performance of the TASER 10™ system within 18 months of its introduction into service.

124. Recommendation 12

SACMILL has not been made aware of any Child Rights Impact Assessment (CRIA) undertaken in connection with the TASER 10™ system.³⁷ SACMILL advises that this must now be undertaken as the UK Government has made a public commitment to give due consideration to the United Nations Convention on the Rights of the Child (UNCRC) when making new policy or legislation. This TASER 10™ statement may require revision subject to the outcome of the CRIA. Additionally, SACMILL emphasises that children who are encountered as offenders, or alleged offenders, are entitled to the same safeguards and protection as any other child and due regard should be given to their safety and welfare at all times.³⁸

³⁷ <https://www.unicef.org/childrightsandbusiness/media/541/file/Childrens-Rights-in-Impact-Assessments.pdf>

³⁸

https://assets.publishing.service.gov.uk/media/6849a7b67cba25f610c7db3f/Working_together_to_safeguard_children_2023_-_statutory_guidance.pdf

125. Recommendation 13

Should the system be authorised, it is vitally important that any significant deviations from the medical implications articulated in this statement are reflected in a revised statement.

Furthermore, should any substantive element of the TASER 10™ system change, SACMILL must be informed and any consequent medical implications considered.

A handwritten signature in blue ink, appearing to read 'J Payne-James', with a long horizontal flourish extending to the right.

Professor Jason Payne-James

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Chair of SACMILL

28 July 2025